

Lab 05 – Op Amp Circuits I (DAC)

Abdon Morales

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Description

In this lab, we built and tested a 4-bit digital-to-analog converter (DAC) based on an op-amp R-2R ladder. The ladder takes a 4-bit binary input set by switches S_1 through S_4 . Each bit connects its rung either to the full-scale voltage V_{FS} (logic 1) or to ground (logic 0), and the inverting op-amp stage turns that input into an analog output voltage. The ideal output follows

$$V_{out} = -(\text{Binary Value}) (2^{-N}) V_{FS},$$

so each bit adds a negative, binary-weighted step to the output, with B_4 as the MSB and B_1 as the LSB.

We first simulated the circuit in LTSpice with an ideal op-amp, once using the proper ratio (50 k Ω and 100 k Ω) and once using a modified ratio made from accessible parts (47 k Ω and 100 k Ω). We then built the circuit on the XK700 trainer with a 741 op-amp and measure the output voltage for every input from 0000b to 1111b.

Procedure

1. Verified the component values before building: four 47 k Ω resistors, five 100 k Ω resistors, and one 741 op-amp.
2. With the XK700 power off, built the 4-bit R-2R ladder DAC on the breadboard following the prelab schematic and the 741 pinout.
3. Wired the four bit inputs to the SW1–SW4 data switches, where down is LOW (0 V) and up is HIGH (5 V), so each switch selects logic 0 or logic 1 for its bit.
4. Connected the ground points to GND on the power supply, and connected +12 V and –12 V from the supply to V_+ (pin 7) and V_- (pin 4) on the 741.
5. Double-checked every connection, then powered up the trainer.
6. Stepped the switches through each binary value from 0000b to 1111b and recorded the output voltage V_{out} for each.

Simulations

The goal of the simulations is to verify that the output follows the DAC equation. The resolution (least-significant bit) is set by the number of bits:

$$R = \text{LSB} = V_{FS} (2^{-N}),$$

which for $N = 4$ and $V_{FS} = 5$ V gives an LSB of $5/16 = 0.3125$ V. The analog output for any input combination is

$$V_{out} = -(\text{Binary Value}) (2^{-N}) V_{FS}.$$

Two simulation runs are required, both using `.include opamp.sub` with the ideal `opamp` component:

- Proper ratio: $R = 50\text{ k}\Omega$, $2R = 100\text{ k}\Omega$.
- Modified ratio: $R = 47\text{ k}\Omega$, $2R = 100\text{ k}\Omega$.

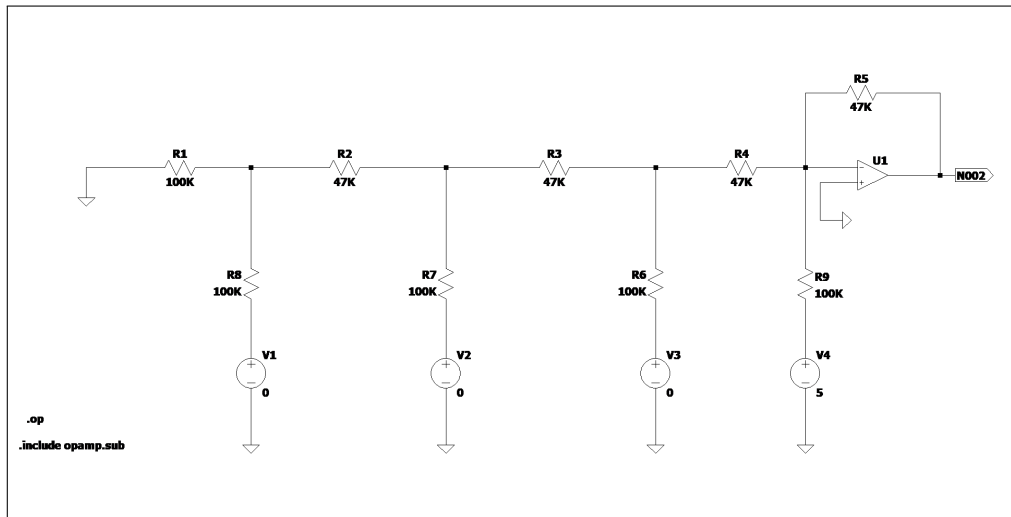


Figure 1: LTSpice schematic

```
* C:\Users\Abdon Morales\OneDrive - The University of Texas at Austin\
  Summer 2026\ECE 402\Lab Notes\Lab 05\Cirrcuits Lab05.asc
* Generated by LTspice 26.0.2 for Windows.
R1 N003 0 100K
R2 N004 N003 47K
R3 N005 N004 47K
R4 N001 N005 47K
R5 N002 N001 47K
R6 N008 N005 100K
R7 N007 N004 100K
R8 N006 N003 100K
XU1 N001 0 N002 opamp Aol=100K GBW=10Meg
R9 N001 N009 100K
V1 N006 0 0
V2 N007 0 0
V3 N008 0 0
V4 N009 0 5
.include opamp.sub
.op
.backanno
.end
```

Listing 1: LTSpice netlist

The proper-ratio run ($50\text{ k}\Omega/100\text{ k}\Omega$) reproduced the ideal equation exactly. The modified-ratio run ($47\text{ k}\Omega/100\text{ k}\Omega$) deviated from the ideal: the lower bits came out slightly larger in magnitude

than ideal, while the upper bits fell short, so the full-scale output (1111) reached only -4.4802 V instead of -4.6875 V .

Table 1: Simulation run results, V_{out} (V), $V_{FS} = 5\text{ V}$.

$B_4B_3B_2B_1$	Decimal	Proper (50k/100k)	Modified (47k/100k)
0000	0	0.0000	0.0000
0001	1	-0.3125	-0.3164
0010	2	-0.6250	-0.6139
0011	3	-0.9375	-0.9303
0100	4	-1.2500	-1.1999
0101	5	-1.5625	-1.5163
0110	6	-1.8750	-1.8138
0111	7	-2.1875	-2.1302
1000	8	-2.5000	-2.3500
1001	9	-2.8125	-2.6664
1010	10	-3.1250	-2.9639
1011	11	-3.4375	-3.2803
1100	12	-3.7500	-3.5499
1101	13	-4.0625	-3.8663
1110	14	-4.3750	-4.1638
1111	15	-4.6875	-4.4802

Measurements

Table 2: DAC output for each 4-bit input ($V_{FS} = 5\text{ V}$). Ideal $V_{out} = -(\text{decimal})/16 \times 5\text{ V}$.

$B_4B_3B_2B_1$	Decimal	Ideal V_{out} (V)	Measured V_{out} (V)	% Error vs Ideal
0000	0	0.0000	0.000	—
0001	1	-0.3125	-0.318	1.76
0010	2	-0.6250	-0.615	1.60
0011	3	-0.9375	-0.931	0.69
0100	4	-1.2500	-1.203	3.76
0101	5	-1.5625	-1.519	2.78
0110	6	-1.8750	-1.816	3.15
0111	7	-2.1875	-2.132	2.54
1000	8	-2.5000	-2.335	6.60
1001	9	-2.8125	-2.650	5.78
1010	10	-3.1250	-2.948	5.66
1011	11	-3.4375	-3.263	5.08
1100	12	-3.7500	-3.536	5.71
1101	13	-4.0625	-3.852	5.18
1110	14	-4.3750	-4.149	5.17
1111	15	-4.6875	-4.464	4.77

Observations & Discussion

Our measured outputs followed the right stair-step pattern, but they did not match the ideal R-2R values. The low codes came out slightly high in magnitude (0001b measured -0.318 V against an ideal -0.3125 V), while the upper codes came out low, with the biggest gap at the MSB (1000b measured -2.335 V against an ideal -2.500 V, a 6.6% error). The error stayed above 5% across most of the upper half of the range. This is a ratio problem, not a random measurement problem. A true R-2R ladder needs R to be exactly half of $2R$, but we built it from 47k and 100k, so the ratio was 0.47 instead of 0.50. That changes how current divides at each node. The bits do not all shift by the same amount or even in the same direction, which is the signature of a ratio error rather than a simple gain or offset.

The measured values lined up closely with the modified 47k/100k LTSpice run. Every code agreed within about half a percent (1000b measured -2.335 V against a simulated -2.350 V, and full scale 1111b measured -4.464 V against -4.4802 V). So the SPICE model predicted the real circuit well once it used the actual resistor values. This matters because it pins the error on the design itself. The model and the bench agree, and both sit away from the ideal for the same reason, the 47k to 100k ratio.

We also have to consider human error in building the circuit as a source of discrepancy. A rung wired to the wrong node, a resistor placed in the wrong position, a loose breadboard contact, or a misread on the meter would all show up as extra error on top of the ratio problem. The close agreement with the simulation actually argues that our build was mostly correct, because a real wiring mistake would have thrown specific codes far off instead of leaving every code within half a percent of the predicted value. The small differences that remain are about what we would expect from resistor tolerance, contact resistance on the breadboard, and reading the meter, rather than from a construction mistake. Even so, rechecking each connection against the schematic and verifying every resistor value before powering up is the main way we keep that source of error out.

The obvious question is whether the feedback resistor can correct this. It cannot, at least not fully. In this inverting setup the output is $V_{out} = -R_f \sum I_{bit}$, so R_f scales every bit by the same factor. It is a single gain knob. If we size R_f to make the MSB exact, we would scale the output by $2.5/2.335$, but that same factor applied to the LSB, which was already high, pushes it further off. Because the bit errors point in opposite directions, no single multiplier fixes all of them at once. The most R_f can do is trim one code, usually full scale, and leave the rest off.

The real fix is to rebuild the ladder so the 2:1 ratio is exact instead of approximate, using the resistor values we already have by combining matched parts. We can make each R from two 100k resistors in parallel, which gives exactly 50k, and keep the $2R$ rungs as single 100k resistors. The ratio is then 50k:100k by construction. Building the ratio out of identical resistors is far more accurate than trusting 47k against 100k, because parts from the same batch track each other, so the ratio error drops to their matching rather than their absolute tolerance. The same idea works the other way, making each $2R$ from two 47k resistors in series for 94k and keeping R as a single 47k. Either approach should bring the ladder inside the 0.5% target. The tradeoff is that it needs more resistors than the basic build.